

HENOK WEHIBE

London, United Kingdom

henokwehibe@gmail.com | linkedin.com/in/henokwehibe | github.com/micrometre | henok.cloud

WORK EXPERIENCE

MicrometreUK

Lead Developer / Sole Engineer

2022 – Present

Designed, built, and operate MicroANPR — a complete Automatic Number Plate Recognition ecosystem spanning a FastAPI processing backend, Ansible-managed bare-metal infrastructure, a Tauri 2 cross-platform operator client, and a Flutter edge-AI mobile camera app. Achieved 97%+ recognition accuracy on UK plates; system audited against ISO/IEC 27001, 27701, 25010, 12207, and 30141.

- **Backend API & Video Processing (fastapi-anpr-api):** Engineered async Python/FastAPI REST API with SQLAlchemy ORM and PostgreSQL. Processes uploaded videos frame-by-frame with OpenCV and TensorFlow Hub models (Faster R-CNN for detection, CCT-based OCR for character recognition). Serves multi-format exports (JSON, CSV, PDF, ZIP), WebSocket real-time progress, and a resident/visitor permit CRUD system with CSV bulk import. Added Progressive Web App support with Service Worker offline caching.
- **Computer Vision & ML Model Pipeline (training, model_weights):** Built custom TFLite model toolchain: scripted download, quantisation (uint8 EfficientNetB0), TFLite metadata embedding, and label file generation. Evaluated Intel OpenVINO GPU-accelerated inference (MobileNet SSD) with Jupyter-based CPU-vs-GPU benchmarking on Intel Iris Xe. Maintained PaddleOCR and PaddleDetection pipelines for UK plate formats.
- **Edge AI Mobile Camera (anpr-camera — Flutter / Dart):** Developed real-time on-device vehicle detection app using Google ML Kit Vision with custom TFLite classifier running inference on raw NV21/BGRA8888 camera frames. Implemented MVVM architecture (Provider state management), hardware-encoder-safe video export with even-dimension rounding, and Android Scoped Storage (MediaStore API) for zero-permission file sharing.
- **Cross-Platform Operator Client (anpr-client — Tauri 2 / Rust):** Built desktop and mobile client with Tauri v2 (Rust backend, HTML5/Tailwind frontend). Enforced strict network sandboxing via Rust capability scopes (capabilities/default.json allow-list). Automated cross-compilation: Linux .deb/AppImage, Windows .exe (MinGW-w64 + NSIS cross-build from Ubuntu), and Android .apk. Integrated client-side Excel (SheetJS), PDF (jsPDF), and CSV export engines.
- **Infrastructure-as-Code & Security (fastapi-anpr-infra — Ansible):** Authored Ansible playbooks for full bare-metal provisioning: SSH hardening (PermitRootLogin no, key-only auth, AllowUsers), Fail2ban jails monitoring Docker Nginx logs, non-root container isolation via APP_UID/APP_GID build-args, automated local CA/TLS certificate generation (mkcert), systemd daemon registration with boot auto-start, and one-command backup/restore of PostgreSQL data volumes.
- **Go Microservices & Real-Time Streaming:** Implemented Go/Gin SSE endpoints and Redis Pub/Sub for zero-latency plate detection push notifications. Built Next.js dashboard with real-time data visualisation and tariff calculation for parking operators.

Projects

2020 – Present

1. **Intel OpenVINO GPU Inference Toolkit** — Production-ready object detection on Intel Iris Xe with OpenVINO 2024+. Automated driver/runtime setup scripts for Ubuntu 26.04 LTS; Jupyter notebook benchmarking CPU vs GPU (up to 4× speedup). MobileNet v1/v2 SSD model zoo integration.
2. **CLI Agent** — AI-powered terminal assistant using local LLM models (llama.cpp / OpenVINO backend) for shell command execution and automated README generation.
3. **Browser-Based AI Chat (WebLLM)** — 100% client-side chat app using WebLLM and WebGPU for hardware-accelerated AI inference with zero server dependencies.
4. **Daily Cashier** — Offline-first sales management system with Next.js, SQLite WASM, and OPFS (Origin Private File System) for persistent client-side storage.

5. DevOps & Infrastructure

- Ansible roles for automated WireGuard VPN deployment; Kubernetes clusters (kubeadm, MicroK8s) with Calico/Flannel.
- Custom Debian images via Packer; private Docker Registry and Gitea on AWS, GCP, and Azure.
- Nginx/HAProxy reverse proxy with TLS termination, HSTS, and Fail2ban intrusion defence.

Freelance/Self Employed

Full Stack Web Developer

2020 – 2022

- Built e-commerce sites, landing pages, and custom web apps for clients using React, Next.js, and WordPress.
- Delivered end-to-end hosting solutions: domain registration, Nginx/SSL configuration, Cloudflare/Vercel edge deployment.
- Conducted performance, accessibility, and SEO audits with measurable improvements.

TECHNICAL SKILLS

Languages	Python, Go, Rust, Dart, JavaScript/TypeScript, SQL, Bash
Backend	FastAPI, Gin (Go), Flask, Node.js, REST APIs, WebSockets, SSE
Frontend	React, Next.js, Astro, Flutter, Tauri, HTML5, Tailwind CSS
AI/ML & Vision	TensorFlow / TF Hub, TFLite, Google ML Kit, OpenVINO, OpenCV, PaddleOCR
Databases	PostgreSQL (SQLAlchemy, Alembic), Redis, MySQL, MongoDB, SQLite WASM
DevOps & Cloud	Ansible, Docker, Kubernetes, Nginx, Fail2ban, Systemd, AWS, GCP, Cloudflare
Tools	Git, Make, Packer, Vagrant, FFmpeg, MinGW, NSIS, mkcert

CERTIFICATIONS

FreeCodeCamp	Responsive Web Design
FreeCodeCamp	JavaScript Algorithms and Data Structures

EDUCATION

South Thames College

London, UK

Diploma	Information Technology
GCSE	Triple Science (Biology, Chemistry, Physics), English, Maths